

## The next step

With the advent of cloud computing as an infrastructure choice, the evolution wheel is again turning, with companies beginning to investigate the technology options available to them. Private cloud solutions —whether on-premise in your own data centre or off-premise (hosted by a partner) — could provide even greater possibilities for performance with enhanced IT automation, elasticity of resources, and deeper capabilities for tracking usage.

An important stage has been reached in the evolution of this technology with the virtualising of mission critical applications, as well as the transformation that has resulted in virtualised resources becoming a pool. At this point, infrastructure is shared and IT must embrace a service management focus to deliver the private cloud in which physical and logical resources are made available through a virtual service layer across the enterprise. Whether an organisation selects embedded or third-party management applications, one thing is certain—implementing better control over business virtual machines will give it more flexibility to take virtualisation to the next stage.

Anyone already using a virtualised server environment providing basic IT elasticity and scalability will be aware that organisations are able to enjoy the extra advantages provided by private cloud implementations such as self-servicing, increased automation and measurement, and the delivery of IT Infrastructure as a Service (IaaS). With cloud computing, workloads are allocated to software and services, which are utilised over a network of servers in various locations, in a group known as a cloud. These are assigned to connections, and accessed over a thin client or other access point, like an iPhone or laptop. Users can access the cloud for resources on demand.

## The latest happenings

- **Performance**
  - Recent advances in hardware and software have removed some of the performance concerns associated with virtualisation
  - Performance guarantees: If you have a database transaction requirement of a few milliseconds, it is very difficult to provide that guarantee in a virtualised environment.
- **Flexibility**
  - Live VM migration without downtime
  - Fault tolerance
- **Management of virtualised technologies**
  - VM action log
  - VM snapshots
- **Isolating work and home smartphones on a single physical handset**
  - Security

- Ability to isolate data and applications corresponds to another aspect of the technology
- Isolation enforced at the lowest level
- Among organisations currently using virtualisation, business-critical applications are now beginning to come into the limelight. Corporations are most willing to virtualise Web, databases and e-mail applications.
- Enterprises are willing to place sensitive applications like HR, accounting, ERP and CRM into a virtualised/cloud environment.
- Security, disaster recovery and governance are road blocks in virtualisation adoption.
- When it comes to actually implementing virtualisation and cloud computing solutions, organisations often discover it a complicated encounter. Actually, most of them discover it necessary to solicit the help of a third party to be able to completely apply their solution.
- According to Gartner, the overall virtualisation market reached a healthy \$2.7 billion in 2011 and it is expected to grow to \$6.3 billion by 2014.
- The price of handling virtual environments may be higher than predicted because, as usage improves, the price of handling hosts may stay the same or increase as companies come to grips with technological innovations they haven't formerly dealt with.
- In order to truly make use of virtualisation, push up the amount of virtualised hosts and rely on the technological innovations to assist mission-critical applications, corporations must modify their virtualisation techniques to account for the changes this technological innovation brings to the hosting server room. 

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